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The Automation Arms Race in the Digital labour Market

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Abstract

This paper examines how automation and generative AI affect signalling and matching in digital labour markets. Using LinkedIn as a central example, we study how AI-assisted tools reduce the cost of producing application materials, thereby weakening the informational content of application-level signals. Empirical evidence and descriptive accounts suggest that both applicants and employers increasingly rely on automation, compressing effort costs on both sides of the market. We analyse these developments by adapting Spence's (1973) job-market signalling model to a digital setting and extending it to a cheap-talk framework. The model shows that as signalling costs fall and become more uniform, separating equilibria are harder to sustain, leading to noisier matching outcomes. Overall, the results suggest that while digital platforms initially improved matching efficiency, widespread automation may erode the credibility of traditional signals in digital labour markets.

Contents

1	Introduction	1
2	Conceptual Background	3
3	Literature Review	4
3.1	The shifting in economic costs	4
3.2	LinkedIn as a digital signalling environment	5
3.3	Automation, AI, and the erosion of signals	6
3.3.1	Theoretical extensions of signalling and cheap talk	7
4	Digital labour market - LinkedIn	9
4.1	AI, automation, and paid visibility	10
5	Spence's Signalling Model	11
5.1	Model Theory	11
5.2	Theoretical Validation of Spence's Signalling Model in Digital Labour Market	13
5.3	Automation-Induced Cheap Talk Model in Digital Labour Markets	14
5.3.1	Equilibrium with arbitrary number of intervals	17
6	Consequences of automation arms race among heterogenous candidate pool	18
6.1	Threshold of bias in automation induced labour market	19
6.1.1	Summary	20
7	Discussion	21
7.1	Interpretation of Results	21
7.2	Implications for Efficiency, Fairness, and Market Design	21
7.3	Limitations and Future Work	22
8	Bibliography	23

1 Introduction

Hiring under uncertainty is a central concern in labour economics, as employers must form beliefs about applicants' productivity before they can observe it, leading to potential mismatches and information frictions. Michael Spence's (1973) seminal theory of job-market signalling provides a framework for this challenge: workers indicate ability through costly signals, while employers interpret these signals to form expectations about productivity. Because these signals are differentially costly across types, they act as credible sorting mechanisms, allowing firms to minimise mismatches while workers balance the costs and benefits of signalling (Gibbons, 1992, [9]).

However, in modern labour markets, much of this interaction has shifted toward digital platforms, with one major platform being LinkedIn. This platform allows candidates to display (signal) their skills, education, work experience, and endorsements for the free disposal of both peer candidates and employers, who can then observe, evaluate, and even rank applicants algorithmically. In recent years, these interactions have changed even further with the entrance of generative artificial intelligence (AI), which offers tools to create ready-to-use material for job applicants, such as AI-generated résumés and cover letters, hereby lowering the cost of effort of signalling. These tools may quickly ensure textual alignment with job postings and polished grammar, making it easier for all candidates to present an abundance of equally refined applications. This also applies to the employer side, where AI has reduced the cost of job posting and applicant screening, leading to possibly less informative job descriptions and a greater congestion in the market (Cui et al., 2025 [15]; Wiles & Horton 2025 [24]). In such an AI mediated interaction with minimal effort costs, perhaps the underlying assumptions of Spence's signalling model may no longer hold? A related theoretical framework is provided by Crawford and Sobel (1982), whose cheap-talk model captures these environments where communication becomes less costly and more subject to noise (Crawford et. al, 1982, [6]). However, at the same time, the widespread use of AI tools on both sides of the market raises the possibility that new costly signals may emerge to restore credibility. As both applicants and employers adopt automation tools, their strategic adjustments can generate an "automation arms race" (Thornhill, 2025, [21]). This paper therefore asks how digitization and automation affect the credibility and efficiency of signalling, using the following research questions.

1. *How does the digital labour market, by using LinkedIn, differentiate between high - and low qualified signals?*
2. *Under what conditions may the LinkedIn job market move from a separating equilibrium toward pooling or cheap-talk equilibria?*
3. *How does the automation arms race in LinkedIn help aligning the incentives of employers with a continuum type space of candidates?*
4. *How do different candidate types including heterogeneity in experience, occupation and job type encounter the consequences of the automation arms race?*

We focus on LinkedIn as a case study because it combines elements of a signalling environment, a labour market platform, and a social network, making both sender and receiver information publicly visible. In particular, we adopt the two theoretical frameworks from Spence's job-market signalling model and Crawford and Sobel's cheap talk model (Gibbons, 1992, [9]) to examine how AI and automation may impact the interactions between candidates and recruiters, how these changes influence the credibility of signals, and what implications this has for the efficiency of digital hiring. We will start by section 3 with a literature review, summarizing the sources of literature which motivated the idea of our seminar paper. Following this, we have section 4 which is about LinkedIn and its feature. In section 5, we have our economic models at work that extensively goes on into answering research question 1,2 and 3. The section 6 illustrates the consequences of automation arms race in heterogenous candidate pool which consists of experience level, occupation and job and the threshold of bias to distinguish between the heterogenous pool of candidates. The last section, section 7, covers the overall conclusion, inefficiency in the market and some policy designs as a corrective measure. This section also extends some ideas around future work as the seminar paper restricts to put only the model empirics into work not exclusively the findings of the results around it.

2 Conceptual Background

In this paper, we examine how labour-market mechanisms may be shifting as online information and AI tools become widespread. In the literature, the labour market in an internet context is both described by the terms online labour market and digital labour market. However, these terms are also used in a narrower sense to refer to platform-based environments that connects employers with an online labour force, e.g. the MIT economist John Horton has used both online- and digital labour market in two different papers, to explain the same phenomenon regarding recruitment of jobs that may be done remotely. (Wiles & Horton, 2025, [24], Horton et. al, 2017, [13]). Thus, to clarify, in this paper we use the term 'digital labour market' in the broader sense, referring to the labour-market interactions themselves when they are mediated by the internet, e.g. by the use of the platform LinkedIn, but the work demanded is arbitrary. We chose the phrasing digital, as we found it more accurate, linking to the matching processes being digitised.

Additionally, as we have dived into this subject, we have found the literature on labour markets in a digital context to cover a wide range of possible developments in behavioural dynamics, many of which could be relevant when analysing the signals that appear in job search. However, given the size of this paper, we constrain our focus to the signalling interaction between applicant and recruiter at the initial point of contact: the application itself. We therefore do not consider earlier stages, such as how the design of job postings shapes the applicant pool, or later stages, such as interviews and hiring tests. Thus, in this paper, our analysis focuses exclusively on the interface where applicants create their profiles and applications and recruiters evaluate them. And while discretion and recruiter-specific biases have been shown to affect hiring decisions in practice (Hoffman, Kahn, and Li, 2018, [12]), we abstract from such considerations and assume that recruiters act in line with their firm's objective of identifying the most qualified or productive candidate. By doing so, we can concentrate solely on how information frictions, cost-saving measures, and AI tools shape the recruitment process.

3 Literature Review

This section goes through the literature that will form the foundation for our analysis of signalling in digital labour markets. We begin by understanding how digitalisation has changed economic costs in job search and recruitment. Then we turn to empirical evidence on LinkedIn as a digital signalling environment, which will be supported by the technical aspects of section 4. Finally, we review how recent advances in automation and AI affect the cost and informativeness of application-level signals, motivating the theoretical framework developed in Section 5.

3.1 The shifting in economic costs

With the rise of the internet, labour-market interactions have increasingly moved to digital platforms, where job search, vacancy posting, and recruitment are mediated through digital communication and algorithmic matching. Goldfarb and Tucker (2019) provide a broad economic framework for understanding this transformation, arguing that digital technologies reduce several key costs, including search, replication, tracking, and verification. Amongst these, they highlight that the reduced search costs have driven the emergence of “peer-to-peer” platforms dedicated to facilitating matches, where they argue that the use of the internet may have potentially reduced frictions between workers and firms, meaning an improvement in the efficiency of matching markets (Goldfarb & Tucker, 2019, [10]). Supporting this view, Mang (2012) provides empirical evidence using micro-level data from the German Socio-Economic Panel. The paper compares employees who found jobs through online search channels report higher job satisfaction, better promotion prospects, and greater job security compared to those using traditional search methods (Mang, 2012, [19]).

These changes in economic costs are central to understanding how job matching works in digital labour markets. In particular, the quality of written applications plays a key role in the formation of matches between applicants and employers. Wiles, Munyikwa, and Horton (2023) study this mechanism in an online labour market and distinguish between a ‘signaling view’, where writing quality reflects underlying ability, and a ‘clarity view’, where better writing helps employers interpret applicant information more easily. Using a large-scale field experiment, they show that algorithmic writing assistance improves hiring outcomes without reducing employer satisfaction, suggesting that writing quality directly affects match formation - even when underlying productivity is unchanged. This highlights that changes in the cost of writing can alter not only application effort, but also how effectively applicants are matched to jobs (Wiles etl. al., 2024, [25]).

To formalize these changes in economic costs in the job market, we turn to Michael Spence’s (1973) job-market signalling model. In the classic model, individuals of different productivity levels choose costly signals (typically education) to distinguish themselves, and employers interpret these signals to infer worker types. The equilibrium of matching between applicants and recruiters depends on whether a high-ability type can choose a signal costly enough that a low-ability type would not mimic it. If this

condition holds, an equilibrium exists in which recruiters can reliably act on the signals sent. If not, recruiters may face suboptimal outcomes, hiring workers whose true ability is not accurately conveyed. Crawford & Sobel (1982) extends this framework and argues that when signals are effectively costless or provide no credible information, this changes to ‘cheap talk,’ where the signal only supports intention, but without credibly revealing the applicant’s underlying ability (Gibbons, 1992, [9]). These models were developed long before the internet and therefore relies heavily on assumptions about interactions in a traditional job market. Hereby, it provides a foundational framework for our analysis of the traditional versus the modern job-market and will be further explained and extended in Section 5.

3.2 LinkedIn as a digital signalling environment

Within this broader shift toward digital labour markets, LinkedIn has emerged as a central platform for job search and career development. LinkedIn is multifaceted, and functions as both a social network, a signalling environment, and a marketplace all in one, potentially redefining the labour market structures.

A growing amount of empirical literature studies how the use of LinkedIn affects labour-market outcomes and individual expectations. E.g. Peña et al. (2022) examine how usage of LinkedIn shapes individuals’ career outcome expectations, showing that more active engagement with the platform - such as posting content and expanding professional networks - is associated with higher expected career mobility and more positive perceptions of labour-market opportunities. Hereby, their findings suggest that LinkedIn functions not only as a job-search tool, but also as a signalling environment, where users form beliefs about their prospects based on observable profile features and platform interactions (Peña et. al, 2022, [20]).

Related evidence is provided by Wheeler et al. (2019), who study the effects of LinkedIn-based job readiness training on employment outcomes. Using a field experiment, they show that improving candidates’ ability to present themselves effectively on LinkedIn increases job-finding rates. This indicates that how information is displayed and communicated on the platform plays an important role in shaping recruiter responses at the application stage (Wheeler et. al., 2019, [22]).

While these studies establish the relevance of LinkedIn for job search and signalling, they largely focus on earlier platform features and on human-generated content. They do not examine how more recent technological developments affect the cost or credibility of application-level signals. In our review of the literature on how LinkedIn features shape job matching, we have only found partial explanations: either studies that analyse AI tools in general, or studies that examine LinkedIn in a broader, non-AI context. So far, evidence specifically linking LinkedIn’s AI features to changes in signalling has mainly appeared in journalistic articles and personal testing rather than in systematic academic work. Therefore, in analysing how the signalling environment may be changing, we draw on both academic research papers and descriptive articles to inform our discussion.

3.3 Automation, AI, and the erosion of signals

Recent advances in automation and generative AI suggest that the cost structure underlying digital signalling may be changing. Cui et al. (2025) provide direct empirical evidence of this mechanism using data from the online labour platform Freelancer.com. Their findings suggest that although AI assistance improves textual alignment with job descriptions and hereto enhances the formal quality of cover letters, it simultaneously weakens their informative value: after the adoption of AI, the correlation between cover letter quality and callback probability falls by approximately 51 per cent. Consequently, Cui et al. (2025) implies that employers may begin to rely more heavily on alternative, harder-to-fake indicators such as past work reviews. This paper thus indicates that AI reduces the signalling power of application effort, making it more difficult for recruiters to infer an applicant’s true ability (Cui et. al, 2025 [15]).

This claim is further supported by journalistic evidence suggesting that similar dynamics may be emerging on LinkedIn. In the article, with the suitable name: *“I Applied to 2,843 Roles: The Rise of AI-Powered Job Application Bots”*, Koebler (2024) documents the use of AI-powered job application bots that automatically generate résumés and cover letters, to then submit large numbers of applications on LinkedIn at very low cost. Koebler (2024) likewise notes, that while the resulting application materials appear professionally written, they are also often vague and generic, thereby reducing their informational content. This again suggests that automation can substantially lower the effort required to produce polished applications, weakening the credibility of application-level signals in much the same way as observed by Cui et al. (2025). (Koebler, 2024, [16]).

On the employer side, similar cost reductions may also reshape behaviour. In a recent paper, Wiles and Horton (2025) show that generative AI substantially lowers the cost of creating and posting job advertisements, allowing firms to produce detailed and customised postings with minimal effort. While this can lower costs of recruitment and support easier hiring, it may also contribute to noisier matching environments by encouraging the posting of less defined jobs, and even post jobs with no intention of hiring. As a result of the lowered cost of production of job posts, Wiles and Horton (2025) concludes lower matching efficiency and hereto a potential welfare loss for applicants who apply (Wiles & Horton, 2025, [24]).

As production costs fall for both applicants and employers, application materials convey less information about underlying ability, giving rise to an automation arms race in recruitment. Thornhill (2025) and Koebler (2024) argues that digital platforms and recruitment technologies, such as applicant tracking systems (ATS), have transformed recruitment into a technological contest. Applicants respond by increasingly relying on automation tools, including AI-assisted writing, keyword optimisation, and mass-scale application tools, in order to pass automated screening and avoid being filtered out by these ‘black-box’ systems. Employers, in turn, react by deploying even more restrictive algorithms, automated assessments, and scoring mechanisms, often preventing applications below a certain threshold from ever being reviewed by a human recruiter. While these tools are intended to improve screening efficiency, they

further weaken the link between observed signals and underlying ability. As a result, signalling costs become more uniform across applicants, and the informational value of application-level signals erodes (Thornhill, 2025, [21]; Koebler, 2024, [16]).

Hereby, this literature suggests that while digital platforms such as LinkedIn initially reduced frictions and improved information flow in labour markets, recent advances in AI may erode the informational value of application-level signals by compressing effort costs across applicants. These developments motivate the theoretical analysis in Section 5, where we study how changes in signalling costs affect equilibrium outcomes in digital labour markets.

3.3.1 Theoretical extensions of signalling and cheap talk

Beyond the classical Spence framework, several theoretical contributions extend signalling models to settings with noise, cheap communication, and heterogeneous preferences. These extensions are particularly relevant for digital labour markets characterised by automation and reduced signalling costs. E.g. Backus, Blake, and Tadelis (2019) develop an empirical framework for signalling games by analysing millions of eBay negotiations, showing that round - number pricing conveys information about seller type emphasizing on three equilibrium conditions: sorting, beliefs, and incentive compatibility. In the automation arms race of digital labour markets, similar cheap signals - AI-generated résumés, endorsements, and keyword optimization - mimic this mechanism. As automation reduces signalling costs, digital communication on platforms like LinkedIn increasingly mirrors cheap-talk equilibria, where signals remain partially informative but lose credibility and discriminatory (Backus et. al. 2019 [2]). These extensions will serve as support for our modelling further in section 6.

De Haan, Offerman, and Sloof (2011) extend Spence's (1973) job-market signalling model by introducing noise into the signalling process, showing how imperfect information transmission distorts market outcomes. In their framework, even small amounts of noise can destabilize separating equilibria, while higher noise levels push markets toward pooling equilibria where signals no longer distinguish between high- and low-ability types (De Haan et. al., 2011, [7]).

It is crucial to pin down, building on Crawford and Sobel's (1982) *Strategic Information Transmission* and Green and Stokey's (2007) *A Two-Person Game of Information Transmission*, that the sender observes private information, sends a costless message, the receiver updates beliefs and acts; informativeness rises with preference alignment and falls with misalignment; equilibria are partitions (intentional vagueness), with babbling always available. Because messages don't change payoffs directly, multiple equilibria are endemic and standard refinements can't select among them, making experiments crucial for selection principles (Crawford and Sobel, 1982, [6]; Green and Stokey, 2007, [11]).

However, in *Sorting through Cheap Talk: Theory and Evidence from a Labor Market* (Horton, Johari & Kircher, 2024[14]) empirical grounding for the idea that heterogeneous preferences and credible communication channels are critical to maintaining efficiency in algorithmically mediated hiring. This

improvement in sorting efficiency is direct evidence that heterogeneous preferences were present and that communication channels allowed the market to express them. In turn of events, cheap but credible communication from employers, can counteract the inefficiencies of automated sorting, especially when preferences are diverse.

In sum, this framework positions the digital labour market as a setting where digitization reshapes signalling costs. The next section examines the functions of the digital platform in question, LinkedIn, to assess how they influence or disrupt the signalling game. Afterwards, we present a detailed explanation of Spence's job-market signalling model, first in its traditional form and then adapted to the digital labour market.

4 Digital labour market - LinkedIn

This section outlines LinkedIn’s key functions as a basis for analysing signalling and matching dynamics in the digital labour market, focusing on platform features that shape applicant-recruiter interactions at the application stage.

LinkedIn was launched as a start-up in 2003 in the United States and was initially created as a platform to facilitate professional networking. With a slow, but steady growth, it reached one million users within its first year and became profitable after three years. Today, these numbers have increased substantially, reporting figures of more than 850 million users across 200 countries and reported a net revenue of 15.15 billion USD in 2023 (Wikipedia [23]; LinkedIn [18]). Functioning as both a professional social network and a recruitment tool, LinkedIn has become a central platform within the modern digital labour market, shaping how interactions and hiring processes are conducted. Reports show that 72 % of Danish companies use LinkedIn to post job vacancies, while 75% of recruiters report using the platform to conduct background research on potential employees (Ballisager, 2025, [3]).

As the platform has expanded, LinkedIn has developed into a comprehensive digital labour-market ecosystem offering a wide range of services to both job seekers and recruiters. These services range from free tools that allow users to create and maintain professional profiles, to more advanced, subscription-based products that enhance visibility and search capabilities. This development is also reflected in how professional organisations increasingly frame LinkedIn as an essential part of working life. For example, the Danish labour union Djøf explicitly emphasises the central role of LinkedIn in professional careers. In its official guide, Djøf describes LinkedIn as “a key tool in a career where relations, knowledge and visibility play an increasingly important role” (Djøf, 2024, p. 3 [8]), and recommends that members maintain an active presence on the platform. According to Djøf, LinkedIn is more than a digital address book: it allows users to make professional competences visible, engage with relevant networks, and potentially “open new doors” in their careers (Djøf, 2024 [8]). In this sense, LinkedIn has become an important site for how professional identity and employability are expressed in a digitalised labour market.

At its core, LinkedIn allows users to create professional profiles displaying education, work experience, skills, endorsements, recommendations, and network connections. In addition, the platform makes certain user activities observable, such as posts, comments, likes, and shared content. Features such as LinkedIn Skill Assessments, verified certificates, degrees linked to accredited institutions, and published articles or projects further serve as tools through which users can signal competence and experience to recruiters and peers. These profile features resemble application-level signals in traditional labour markets, but are embedded in a publicly observable digital environment, where the applicant can reach out to potential employer by few clicks. As shown in the literature on LinkedIn use, such observable characteristics shape both recruiter responses and users’ own expectations about career opportunities (Peña et al., 2022 [20]; Wheeler et al., 2019 [22]).

4.1 AI, automation, and paid visibility

Beyond its basic functionalities, LinkedIn offers a range of premium services that allow users to further manage visibility and information flow on the platform. On the applicant side, subscription plans such as *Career Premium* provide access to features including information on who has viewed one's profile, the ability to send *InMail* messages outside one's immediate network, job match insights, and advanced job search filters. Premium users also gain access to LinkedIn Learning courses and, more recently, AI-based tools designed to optimise profile content and written application materials. On the recruiter side, services such as *Recruiter Premium* allow firms to search, filter, and rank candidates more efficiently. According to LinkedIn's own reports, the platform claims to reduce hiring costs by up to 50% and time-to-hire by 29% (LinkedIn, [18]). While such figures should be interpreted cautiously, they indicate how LinkedIn positions itself as a cost-saving intermediary in recruitment.

As described by Djøf (2024), premium features are explicitly marketed as tools that make users "more visible and targeted" in the job search process. While these features do not directly alter a user's underlying skills or productivity, they may affect how frequently and prominently a profile is shown to recruiters. This introduces a distinction between signals that reflect professional ability and signals that primarily enhance exposure.

While LinkedIn has increasingly incorporated AI-related features through its own subscription packages, users can also rely on external AI tools that operate alongside the platform. These tools further reduce the cost of producing applications and maintaining an active professional presence at a near-zero marginal effort. Koebler (2024) documents one such case through the use of an open-source, GitHub-hosted AI application that interfaces with LinkedIn's *Easy Apply* system. By combining a large language model with automated browser actions, the program generates resumes and cover letters tailored to individual job postings and submits applications at scale. Once initial parameters are set, the tool can continuously apply to vacancies with minimal human input, effectively automating both the writing and submission process (Koebler, 2024, [16]). Beyond this application automation, AI tools are also increasingly used to manage one's continuous activity on LinkedIn profiles. Lamaj (2025) describes a range of chargeable AI assistants designed to generate profile content, including fully AI-written personal posts, automated engagement through comments and reactions, and optimisation of profile descriptions. These tools are marketed as ways to maintain visibility and signal activity on the platform without requiring much personal effort. Hereby, combining all of these AI tools, this allows users to outsource not only application writing but also continuous profile maintenance and interaction. (Lamaj, 2025, [17])

5 Spence's Signalling Model

To analyse how signalling may change in the setting of the digital labour market infused with AI-tools, we first consider how it was pre-AI. For this, we make use of the job-market signalling framework developed by Michael Spence (1973), which reflects how signalling operated in the traditional labour market. Here, we will make use of the version of Spence's model as it is in the textbook *A Primer in Game Theory* (1992) by Robert Gibbons.

To fit into our focus point, we make certain modifications of the model. In contrast to the original model where education serves as the main signalling device, we assume here that all applicants share an identical educational backgrounds. Consequently, education no longer serves as a basis for differentiation; instead, the recruiter's evaluation is determined by the quality of the job application itself. In this setting, the signal emerges from the applicant's costly effort e in preparing and submitting their application. This effort reflects the applicant's underlying productivity and functions as an observable, though imperfect, indicator of their ability. This modification aligns with arguments presented by Wiles et. al (2024), who highlights that application-writing quality has traditionally functioned as a key costly signal of productivity prior to the widespread use of AI (Wiles et. al, 2024, [25]).

5.1 Model Theory

The model involves two kinds of players: a *sender* (the applicant) and a *receiver* (the recruiter). The applicant seeks to communicate their productivity level by sending a costly signal, while the recruiter must interpret productivity from the signal and decide whether to offer a job interview. However, we assume a market of perfect competition with countless participants, so the game may be repeated countlessly.

A key feature of the signalling environment is asymmetric information. Applicants privately know their ability level, while recruiters do not. Instead, the recruiter observes only the quality of the job application, represented by effort e . Thus, the game contains incomplete information about types but perfect information about actions. Each applicant's ability is exogenously drawn by 'Nature' and denoted by $\eta \in \{L, H\}$, where L corresponds to a low-ability type and H to a high-ability type. This type is privately known only to the applicant. All applicants exert the effort e when preparing their application, hereby incurring a cost $c(\eta, e)$ that is strictly increasing and convex in e . The cost is interpreted as a non-monetary or psychological cost (Gibbons, 1992 [9]).

On the recruiter side, they wish to optimise profit, π which is dependent on the output y and the 'wage' w , yielding: $\pi = y(\eta, e) - w$, where output is dependent on the type of candidate hired. At this stage, the recruiter must choose whether to invite the applicant to an interview, so we let $w \in \{0, 1\}$ denote the outcome of this decision, where $w = 1$ represents a callback and $w = 0$ a rejection. Traditionally w denotes a wage offer in Spence's model, but here it captures the benefit of advancing in the hiring process. Because screening occurs a cost of effort for the recruiter, they only do callbacks for

selected candidates. Hereby, after observing effort e , the recruiter forms a posterior belief, e.g. $\mu(H | e)$ about the applicant's type, so the callback offer can be seen as the function below, which takes in the fraction of hiring the high ability type μ and low ability type $1 - \mu$, hereby:

$$w(e) = \mu(H | e) y(H, e) + [1 - \mu(H | e)] y(L, e)$$

The applicant seeks to maximize their utility, which depends positively on receiving a callback (w) and negatively on the cost of effort. The callback payoff is effort-dependent, meaning $w(e)$, which also follows the structure of the classical model:

$$U_\eta(e) = w(e) - c(\eta, e),$$

To understand the applicants incentives, we first consider the scenario of complete information, where the recruiter directly observes η . In this case, signalling is unnecessary, and each applicant type simply chooses the effort level that maximises net productivity:

$$e_\eta^* = \arg \max_e [y(\eta, e) - c(\eta, e)].$$

In this scenario, the applicants utility would each be assigned the payoff related to their type, yielding:

$$\text{Low-ability} \quad U_L = y(L, e_L^*) - c(L, e_L^*)$$

$$\text{High-ability:} \quad U_H = y(H, e_H^*) - c(H, e_H^*)$$

A crucial assumption of the signalling model is that the marginal cost of effort is higher for the low-ability type than for the high-ability type:

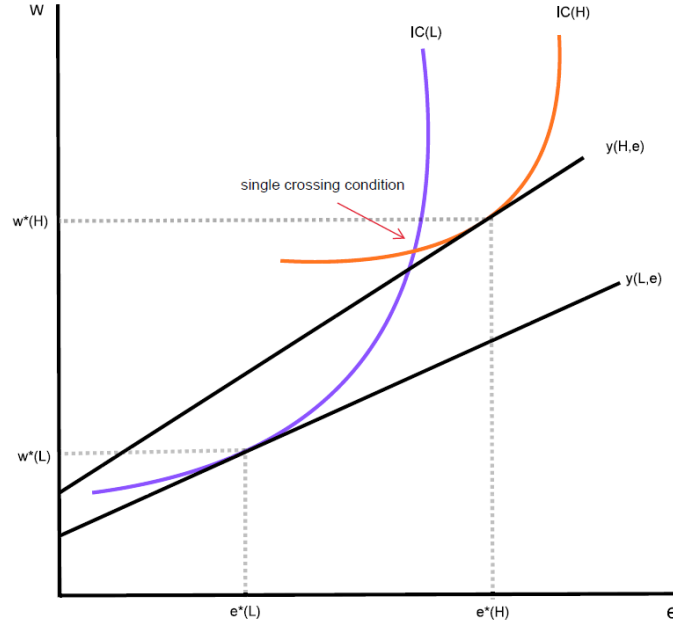
$$\frac{\partial c(L, e)}{\partial e} > \frac{\partial c(H, e)}{\partial e}, \quad \text{for all } e. \quad (1)$$

Combined with $y(H, e_H) > y(L, e_L)$, this implies that high-ability applicants can credibly signal their type by choosing effort levels that are too costly for low-ability applicants to mimic. Thus in complete information there will occur a separating equilibrium.

This equation 1 ensures what is known as the single-crossing property: the indifference curves of high- and low-ability applicants intersect only once. Hereby high-ability applicants therefore have flatter indifference curves, meaning that higher effort levels serve as credible signals of higher productivity. This is illustrated in Figure 1 (Gibbons, 1992 [9]).

The relation of eq. 1 may still apply under asymmetric information, allowing for a continuous separating equilibrium, where high- and low-ability applicants choose different effort levels, enabling the recruiter to infer the applicant's type from the observed signal. A pooling equilibrium would arise if both types chose the same effort, making inference impossible, but under condition 1 such a strategy may be too costly for the low-ability type.

Figure 1: Indifference curves for the high- and low-ability applicants



The distinction between these two equilibrium outcomes is thus central to the signalling model and determines whether the signal - here, the quality of the job application - has any informational value. (Gibbons, 1992 [9])

The game unfolds in three periods:

- (1) Nature draws the applicant's type $\eta \in \{L, H\}$.
- (2) The applicant observes η and chooses to send a signal by effort level e .
- (3) The recruiter observes e , updates beliefs, and decides whether to issue a callback.

5.2 Theoretical Validation of Spence's Signalling Model in Digital Labour Market

Building on the structure of the signalling game outlined in Section 5.1, we now characterise the equilibrium conditions under which signalling remains credible in a digital labour market. Following the framework used by Backus, Blake, and Tadelis (2019), a separating equilibrium requires three components to hold simultaneously: (i) different types must choose different effort levels (sorting), (ii) recruiters must update beliefs rationally after observing effort (belief consistency), and (iii) each type must prefer its own effort choice over mimicking the other type (incentive compatibility).

These conditions ensure that effort retains informational value even when applications are digital and potentially influenced by automation. We outline each condition below, which unfolds over the three periods.

(i) Sorting: In a separating equilibrium, different types choose different effort levels. High-ability applicants optimally select a higher effort level because they face lower marginal costs and obtain a greater benefit from being perceived as high type. Formally, separation requires

$$\arg \max_e [w(e) - c(H, e)] \neq \arg \max_e [w(e) - c(L, e)].$$

In a digital labour market, this corresponds to high-ability candidates investing in stronger or more credible signals (e.g., curated portfolios, verified certificates), while low-ability candidates rely on weaker or low-cost signals.

(ii) Belief Consistency: Recruiters must update beliefs $\mu(s)$ rationally after observing effort. These can be validated if the probability of callback or interview increases monotonically with the posterior belief that a candidate is high ability. Let $\mu(H | e)$ denote the posterior belief that the applicant is high type, then the Bayesian equilibrium for beliefs is:

$$\mu(H | e) = \frac{q f(e | H)}{q f(e | H) + (1 - q) f(e | L)},$$

where q is the prior probability of high type. Digital hiring platforms satisfy this condition if higher-effort applications (e.g., higher-quality submissions) increase the inferred probability of high ability.

(iii) Incentive Compatibility: Given the recruiter's belief mapping, each type must prefer its own effort choice to mimicking the other type. In equilibrium:

$$\begin{cases} w(e_H^*) - c(H, e_H^*) \geq w(e_L^*) - c(H, e_L^*) & \text{(High type will not mimic low type)} \\ w(e_L^*) - c(L, e_L^*) \geq w(e_H^*) - c(L, e_H^*) & \text{(Low type will not mimic high type)} \end{cases}$$

This condition fully captures the idea that high types find stronger signals worthwhile, whereas low types find imitation unprofitable given the higher relative cost of effort.

If distinct sorting by type, rational employer belief updating, and incentive-consistent effort differentials hold jointly, then LinkedIn's labour market satisfies the conditions for a separating equilibrium, where credible digital signals continue to differentiate high- and low-ability candidates even amid automation.

5.3 Automation-Induced Cheap Talk Model in Digital Labour Markets

While Spence's signalling model provides the foundation for understanding costly signals in traditional labour markets, it becomes less suited for environments where ability varies continuously and signals are increasingly cheap to produce. Digital hiring platforms, especially with the rise of generative AI, make it far easier for applicants to generate polished and standardised application materials, reducing the effective cost of signalling (Goldfarb & Tucker, 2019 [10]; Wiles & Horton, 2025 [24]; Koebler, 2024 [16]). As signals become cheaper and more homogeneous, the cost structure required for classical signalling weakens, and the informativeness of application-level signals declines.

At the same time, real labour markets feature continuous heterogeneity rather than the binary types assumed in Spence's model. Ability lies along a spectrum - some candidates are slightly above average, others far above, and others slightly below. This continuum shapes both how much signalling effort individuals choose to invest and how easily (or difficult) firms can distinguish between them. To capture communication in such environments, we adopt the cheap-talk framework of Crawford and Sobel (1982), which accommodates continuously valued types and allows for messages that may be only partially informative of their type, depending on the degree of alignment between candidates and recruiters.

In Spence's binary setup, there exists either perfectly separating (clear signals) or pooling (no information). However in reality, as communication technologies evolve or costs of efforts shift, alignment changes smoothly not abruptly. As LinkedIn introduced AI résumé tools, the effort cost of producing polished applications fell slightly. However, it did not instantly destroy all credibility; but messages became a bit noisier. Over time, as more people use automation, the credibility erodes further. This is where we bring Crawford and Sobel (1992[6]) Cheap Talk model that allow the degree of misalignment b to vary continuously, leading to graded informativeness, from fully informative to completely uninformative communication. In the recruitment environment where AI plays a crucial role in evaluating candidates (senders) on digital signals, communication in the form of messages do not have direct implication on the payoff as they used to (Crawford and Sobel, 1992, [5]; Cui et. al, 2025, [15]). Meaning these messages are costless to produce and effects payoffs indirectly when employers (receivers) update their beliefs and action.

In such setting, it is essential that three necessary conditions suffice in a continuous type space which capture the degree of preference alignment with a continuous parameter for the possible existence of informative equilibria. These different candidate types must have different preferences over the employer's actions; so as to say when every candidate quoting "I am qualified" in their application will convey no meaningful message to the employer (Crawford and Sobel, 1992, [5]). Each candidate must send distinct application with motivational goals. The employer must prefer different actions depending on the candidate's type which eventually makes employer care about the messages and they take action based on it; The employer's preferences must not be completely opposed to the candidate's creating room for credibility or otherwise opposing preferences would make the employer perceive untruthfulness in the application of the candidate.

Having identical timing to the traditional signalling game, cheap talk game lays out the below foundation for when automation is induced making only the payoffs differ.

- (1) Nature draws a type t_i for the candidates (senders) from a set of feasible types

$$T = \{t_1, t_2, \dots, t_i\}$$

from a uniform probability distribution $p(t_i)$, where $p(t_i) > 0$ for every i and

$$\sum_{i=1}^I p(t_i) = 1.$$

- (2) The candidate observes t_i and then chooses a costless message x_j from a set of feasible messages

$$M \in \{x_1, x_2, \dots, x_J\}, M = [0, 1]$$

It is imperative to mention that since all candidates use the same distribution, messages are completely uninformative about their type. This makes it weakly optimal for candidates, regardless of t draw m randomly from $p(t_i)$.

- (3) The employer (receiver) observes x_j (but not t_i) and then chooses an action a_k from a set of feasible actions. We will refer the action here is call back for next interview round.

$$A \in \{a_1, a_2, \dots, a_K\}, A = [0, 1]$$

- (4) Hence the payoffs of both candidate and employers respectively are:

$$U_C(t_i, a_k) = -[a - (b + t)]^2 \quad \text{and} \quad U_E(t_i, a_k) = -(a - t)^2$$

where $b = b_{gf}(A) \geq 0$ is a *group-specific bias with heterogeneity in candidate pool* that increases with automation intensity $A \in [0, 1]$:

$$b_{gf}(A) = \beta_{0g} + \beta_{1g}A_g^C + \beta_{2g}A_f^E + \beta_{3g}A_g^C A_f^E$$

where β_{0g} captures baseline noise and β_{1g} measures how sensitive group g 's signals are to candidate's side automation and β_{2g} captures how sensitive they are to employer side automation. For the group g of candidates the automation used is A_g^C and for the group of employers it is A_f^E . Intuitively, we will expect an upward bias β_{1g} from candidate group with automation in application with CVs and cover letters made with AI. On the other hand from the employer side we would find automation in the form of AI screeners, automated ranking systems which raises misalignment β_{2g} and nuances are missed. The interaction term β_{3g} when both automate at once arms race dynamics is captured. This is important to distinguish whether it's a race to the bottom (amplification) or a race to efficiency (alignment).

The candidate's true type space gets divided into intervals or bins. The type space is partitioned into n intervals,

$$(0, x_1), (x_1, x_2), \dots, (x_{n-1}, 1),$$

such that all types within the same interval send the same message, showing partial pooling equilibrium, while types in different intervals send different messages. When $n=1$, pooling or babbling equilibrium occurs as all types send the same messages. In this case, the message conveys no information and the employer just ignores it. For each group g , as $b_{gf}(A)$ increases, the maximum number of informative communication "bins", $n^*(b_{gf}(A))$, shrinks. Communication becomes coarser, and signals convey less information. Only when $b_{gf}(A)$ is near zero, $n^*(b_{gf}(A))$ rises and automation does not distort incentives and detailed, trustworthy communication occur.

5.3.1 Equilibrium with arbitrary number of intervals

In the standard Crawford-Sobel model, the Candidate's type $t \in [0, 1]$ represents true ability, and the Employer's action a corresponds to the employer's evaluation (callback probability). The degree of preference misalignment is captured by $b = b_g(A) \geq 0$, where the Candidate's preferred action is $a_S = t + b$ and the Receiver's is $a_R = t$. Following Crawford and Sobel(1982[6]), let the type space $[0, 1]$ be divided into n equilibrium intervals (steps), $(x_0, x_1), (x_1, x_2), \dots, (x_{n-1}, x_n)$ with $x_0 = 0$ and $x_n = 1$. Each interval corresponds to one message (like "low," "medium," "high," etc.). When the employer gets a message from interval (x_{k-1}, x_k) , the employer forms belief t which is uniformly distributed over that interval. Recalling that the employer's utility is quadratic, her optimal action is to choose the average of the possible values of t which is the midpoint of the interval she believes t lies in, seen as:

$$a_k = \frac{x_{k-1} + x_k}{2}.$$

For candidate's action to be stable, a boundary type ($t = x_k$) must be indifferent between sending the message for the lower interval or the one for the higher interval:

$$U_S(x_k, a_k) = U_S(x_k, a_{k+1}).$$

Thus utility curve of candidates using $U_S = -(a - (t + b))^2$, we have:

$$|a_{k+1} - (x_k + b)| = |a_k - (x_k + b)|.$$

The candidate at $t = x_k$ is exactly at the midpoint between her ideal action $(x_k + b)$ and the employer actions a_k and a_{k+1} .

$$(x_k + b) - \frac{x_{k-1} + x_k}{2} = \frac{x_k + x_{k+1}}{2} - (x_k + b).$$

$$x_{k+1} - x_k = (x_k - x_{k-1}) + 4b.$$

Each successive interval (step) must be $4b$ longer than the previous one. All the intervals must fill up the unit interval $[0, 1] = d + (d + 4b) + (d + 8b) + \dots + [d + 4b(n - 1)] = 1$

$$d = \frac{1}{n} [1 - 2bn(n - 1)].$$

$$n < \frac{1}{2} \left(1 + \sqrt{1 + \frac{2}{b}} \right).$$

Thus modelling the bias with automation-driven heterogeneity with the equilibrium number of bins is:

$$n^*(A) = \frac{1}{2} \left(1 + \sqrt{1 + \frac{2}{b_g(A)}} \right).$$

To extend this empirical evidence one can calibrate the model using different levels of n and b . As n increases, there are more partitions of the type space - meaning communication is richer and more

detailed. As b grows, these bins spread apart (intervals get wider), so the candidate's ideal point drifts further from the employer's midpoint, they become less aligned and communication becomes coarser.¹ It is needless to conclude that the automation will not have uniform bias across the whole labour market. However, it is crucial to find out how the degree of misalignment changes with does automation arms race across different candidate groups.

6 Consequences of automation arms race among heterogenous candidate pool

Automation in hiring reshapes labour-market signalling by reducing the cost and increasing the uniformity of application-level signals such as résumés, cover letters, and online profiles. While these tools lower participation costs, they weaken the link between presentation quality and underlying ability, raising baseline noise in signal interpretation. In our framework, this effect is captured by a higher group-specific intercept β_g , consistent with evidence that screening technologies are most consequential when productivity is difficult to observe ex ante (Autor, 2001[1])

The resulting erosion of signal credibility is heterogeneous across experience groups. Early-career candidates, who lack reputational capital and verified performance histories, rely disproportionately on low-cost, text-based signals that are especially susceptible to automation-induced noise. This implies a steeper candidate-side automation gradient $\beta_1 g > 0$ causing such groups to reach noisy-signalling thresholds at lower levels of automation. In contrast, experienced workers retain access to costlier and harder-to-fake signals - such as references, leadership histories, and demonstrated output - resulting in lower effective bias growth. A recent Stanford scholar study (Brynjolfsson, et.al, 2025[4]) finds that since the widespread adoption of generative AI, early-career workers (ages 22-25) in the most AI-exposed occupations have experienced a 13 percent relative decline in employment even after controlling for firm-level shocks. In contrast, employment for workers in less exposed fields and more experienced workers in the same occupations has remained stable or continued to grow.

Occupational differences further shape how automation affects signal credibility. In technical fields such as engineering, data science, and design, candidate signals are often grounded in demonstrable output-code repositories, portfolios, or measurable project results - which remain resistant to automation-induced noise because producing high-quality output continues to require skill. Automation primarily accelerates the transmission of these signals without substantially degrading their informativeness (Autor, 2001, [1]). In contrast, occupations centered on communication and judgment, including human resources, marketing, and management, rely more heavily on language-based and self-reported signals. As automation

¹A babbling equilibrium will always exist when $n^*(A) = 1$ as there is just one interval which shows the candidate and employer are at their furthest misalignment. When $n^*(A) \geq 2$, the system reaches a cheap talk equilibrium as communication loses informational content if $b_{gf}(A) \geq \frac{1}{4}$ so, below $b > \frac{1}{4}$ informative communication exists. When $n^*(A) \geq 3$, $b_{gf}(A) \geq \frac{1}{12}$ so, below $b > \frac{1}{12}$ informative communication exists

lowers the cost of generating and refining such text-based materials, these signals become increasingly uniform and less informative, prompting employers to substitute toward additional screening mechanisms. This pattern is consistent with evidence that sorting efficiency deteriorates when communication becomes insufficiently informative, leading toward pooling or cheap-talk equilibria (Horton, Johari, and Kircher, 2024, [14])

Remote, digitally mediated jobs are also more exposed to automation-driven changes in hiring and task allocation, amplifying these informational effects (Brynjolfsson et al., 2025 [4]). As automation spreads, employers depend more on algorithmically produced profiles and metrics, which increases noise and drives the system toward a “cheap-talk” equilibrium where most candidates appear interchangeable. Onsite work still allows for richer, non-digital signals - interpersonal interaction, body language, everyday behaviour - which preserve some credibility in evaluation. These interpersonal connections give employers additional context, helping sustain a partially informative equilibrium even in an increasingly automated hiring environment. Thus, we argue that automation reshapes inequality through the differential erosion of credible signalling across experience levels, occupations, and job types. This is an important point of departure to study Dynamic Automation Tipping through group-specific increases in baseline noise and automation sensitivity.

6.1 Threshold of bias in automation induced labour market

We can formalize the dynamic automation arms race as a process in which increasing automation intensity A amplifies the noise or bias in how signals of competence are interpreted across heterogeneous candidate groups g . Each group’s observed informativeness, summarized by $\hat{b}^g$, can be benchmarked against the theoretical thresholds derived from noisy signalling theory (Haan et.al 2011, [7]): (i) babbling (pure noise) when $\hat{b}^g \geq \frac{1}{4}$; and (ii) loss of the n -bin structure when $\hat{b}^g \geq \frac{1}{2n(n-1)}$. These thresholds allow us to classify when signalling within a group ceases to transmit meaningful information.

Empirically, this framework implies that different experience, occupational or job-type subgroups will tip into noisy signalling threshold at different levels of automation intensity (Crawford and Sobel, 1982, [6]). We refer to this mechanism as *Dynamic Automation Tipping*, which is the moment when rising automation pushes a group past a noise threshold, so that signals of ability stop being credible or informative. Groups with higher baseline noise or steeper noise–automation gradients will reach collapse thresholds of automation and babble/cheap talk equilibrium sooner. Hence, high-volatility or low-observability labor segments will experience faster erosion of informative signalling, while stable or highly experienced occupations sustain separation longer.

We assumed that the bias parameter for each group evolves with automation according to

$$b_{gf}(A) = \beta_{0g} + \beta_{1g}A_g^C + \beta_{2g}A_f^E + \beta_{3g}A_g^C A_f^E$$

So the tipping condition is when it hits that threshold:

$$\beta_{0g} + \beta_{1g}A_g^C + \beta_{2g}A_f^E + \beta_{3g}A_g^C A_f^E = \frac{1}{2n(n-1)}.$$

We can derive the automation level at which group g loses its n -bin structure for: The candidate-side

tipping level is given by:

$$A_g^{C,*}(n | A_f^E) = \frac{\frac{1}{2n(n-1)} - \beta_{0g} - \beta_{2g}A_f^E}{\beta_{1g} + \beta_{3g}A_f^E}.$$

The employer-side tipping level is:

$$A_f^{E,*}(n | A_g^C) = \frac{\frac{1}{2n(n-1)} - \beta_{0g} - \beta_{1g}A_g^C}{\beta_{2g} + \beta_{3g}A_g^C}.$$

and the level at which candidate side babbles, given employer automation:

$$A_g^{C,\text{babble}}(A_f^E) = \frac{\frac{1}{4} - \beta_{0g} - \beta_{2g}A_f^E}{\beta_{1g} + \beta_{3g}A_f^E}.$$

The employer side babble , given candidate automation:

$$A_f^{E,\text{babble}}(A_g^C) = \frac{\frac{1}{4} - \beta_{0g} - \beta_{1g}A_g^C}{\beta_{2g} + \beta_{3g}A_g^C}.$$

A central implication of Dynamic Automation Tipping is that informational collapse is unevenly distributed across heterogenous candidate pool. Groups characterized by higher baseline noise, lower observability, or greater exposure to automated screening reach tipping thresholds earlier, reinforcing existing inequalities. This aligns with broader evidence that informational frictions and screening technologies disproportionately disadvantage already opaque groups, contributing to unequal labor market outcomes (Autor, 2000, [1]). Automation, in this sense, reshapes inequality through the differential erosion of credible signalling.

6.1.1 Summary

Overall, our analysis shows that LinkedIn initially functions as a separating signalling environment (Research question 1), but as automation reduces signalling costs and increases noise, the platform gradually shifts toward pooling or cheap-talk equilibria (research question 2). At the same time, the automation arms race illustrates that incentive alignment between employers and a continuum of candidate types is only partial and depends on automation intensity and exposure (research question 3). Finally, we show that these dynamics are not uniform, as different candidate groups reach these tipping points at different stages, implying heterogeneous consequences across experience levels, occupations, and job types (research question 4).

7 Discussion

7.1 Interpretation of Results

On the fundament of our model, our findings suggest that automation reshapes the informational structure of the digital labour market. In the classical job-market signalling model (Spence, 1973 as presented in Gibbons, 1992 [9]), the credibility of signals rests on differential costs of effort between high- and low-ability applicants. However, in digital labour markets such as LinkedIn, the marginal cost of producing high-quality signals has declined dramatically (Goldfarb and Tucker, 2019 [10]). With the arrival of LinkedIn, the cost of information and imitation has also fallen: candidates can easily browse others' profiles, replicate phrasing or layout, and benchmark their own signals against top-performing examples. This replication dynamic further homogenizes visible signals, making even low-ability candidates appear similar to high-ability ones. At the same time, as generative AI tools allow candidates to produce standardized, polished résumés and cover letters at minimal cost, signalling becomes increasingly decoupled from true productivity (Cui et al., 2025, [15]). Consequently, the separating equilibrium described by Spence gives way to partial pooling or even cheap-talk equilibria (Crawford and Sobel, 1982, [6]), where signals lose their discriminatory power.

This decline is gradual rather than abrupt. As automation intensity rises, the cheap-talk model predicts that the number of informative partitions, or “bins,” shrinks (Crawford and Sobel, 1982 [5]), creating coarser communication between candidates and employers. Thornhill (2025) refers to this as an “automation arms race,” in which both sides deploy technology to maintain informational advantage, yet overall signal informativeness declines. The outcome is a digitally mediated labour market increasingly characterized by noisy signals, asymmetric learning, and reduced allocative efficiency.

7.2 Implications for Efficiency, Fairness, and Market Design

The automation arms race introduces a double inefficiency. Candidates invest in signalling technologies - premium LinkedIn accounts, AI résumé builders, and automated keyword optimization - that no longer insure credible information, while employers respond with more complex applicant tracking systems and algorithmic filters (Thornhill, 2025, [21]). This mutual escalation inflates costs without improving matching accuracy, resulting in welfare losses similar to those described by De Haan, Offerman and Sloof (2011) in noisy signalling environments.

Furthermore, the digitalization of signalling exacerbates inequalities in visibility and access. High-ability or well-networked individuals can still rely on verified credentials, professional endorsements, and track records - signals that remain costly and credible - whereas less experienced or digitally disadvantaged candidates are confined to low-cost, easily replicable signals that recruiters tend to discount. This aligns with the findings of Horton, Johari and Kircher (2024), who show that credible cheap talk requires heterogeneous preferences and reliable communication channels; when automation erodes these channels, sorting efficiency diminishes. In short, automation risks widening informational inequality between experienced and early-career candidates. Credible communication is essential for efficient sorting (Horton,

Johari, and Kircher, 2024, [14]), while workers with thinner work histories rely more heavily on easily imitated, text-based signals that are particularly vulnerable to screening technologies (Autor, 2001, [1]). In response to the AI assisted writing which diminished informational value, employers place greater weight on alternative signals, such as past work experience, that are less susceptible to AI (Cui et al., 2025, [15]).

7.3 Limitations and Future Work

This discussion assumes competitive market conditions and rational Bayesian updating by employers, following the traditional signalling framework (Gibbons, 1992, [9]). While our analysis focuses on first-impression interactions and the likelihood of callbacks, this does not guarantee eventual employment. In reality, recruitment typically involves multiple stages - interpersonal interviews, skill assessments, and other evaluations - that serve to uncover an applicant's true abilities. Nevertheless, recognition as a promising candidate in the initial stage remains essential, highlighting the continued importance of resumes, cover letters, and other signals of competence within digital platforms.

The functional form for automation-induced bias is imposed rather than derived from micro-level behavior, and the analysis remains theoretical, without validating the predicted heterogeneous tipping points across occupations, experience levels, or job types. As a result, the paper should be read as a conceptual map of how automation can erode signalling on LinkedIn, rather than as an empirical measurement of its magnitude.

Another aspect of the model, is that it abstracts from network effects: on LinkedIn, social capital and algorithmic visibility serve as additional forms of signalling not captured by the simple effort-cost formulation. Though premium subscriptions enhances your visibility, we cannot infer that the bought visibility can be equivalent to social capital. For future research, it could be interesting to empirically estimate how automation intensity A affects belief updating and the bias parameter $b_g(A)$ across industries. Experimental or field-based studies could also investigate whether human recruiters can reliably detect AI-generated application content, and how this detection ability influences hiring outcomes. With the differences in adoption of AI-generated materials for different applicants, it becomes easier to capture and estimate the effects of automation (Cui et al., 2025, [15]).

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